Mismatch forensics — AshLey_CheetahBus_20260819_115747_0005_b05

Input 481 MB vs resim output 453 MB (rR00060100) · generated 2026-09-22 12:33

1 · Run inventory

Sensor	Input streams	Output streams
FC	DETECTION_STREAM, DOWN_SELECTION_STREAM, RADAR_CAPABILITY_STREAM, VSE_STREAM	DETECTION_STREAM, DOWN_SELECTION_STREAM, RADAR_CAPABILITY_STREAM, VSE_STREAM
FL	DETECTION_STREAM, DOWN_SELECTION_STREAM, RADAR_CAPABILITY_STREAM, RDD_STREAM, VSE_STREAM	DETECTION_STREAM, DOWN_SELECTION_STREAM, RADAR_CAPABILITY_STREAM, RDD_STREAM, VSE_STREAM
RL	DETECTION_STREAM, DOWN_SELECTION_STREAM, RADAR_CAPABILITY_STREAM, RDD_STREAM, VSE_STREAM	DETECTION_STREAM, DOWN_SELECTION_STREAM, RADAR_CAPABILITY_STREAM, RDD_STREAM, VSE_STREAM

2 · KPI results

Sensor	Detection	Alignment	Tracker
FC	ok 65.27%	failed	missing —
FL	ok 12.42%	failed	missing —
RL	ok 13.52%	failed	missing —

3 · Explain-like-I'm-13

A radar shouts and listens for echoes — each echo is a **detection** with a distance (range), a speed (velocity) and a direction (theta/phi). The resim is a **video-game replay** of the same drive. To check the replay is faithful, the KPI plays spot-the-difference in three rounds. **Round 1 — same photo number?** Every radar picture has a scan number; only pictures with the same number can be compared (FL here: 372 / 395 pictures overlap). **Round 2 — same echo pigeonhole?** Each echo carries an RDD box number; only echoes from the same box are compared. **Round 3 — close enough?** The two echoes must agree within 2.5 m, 2.5 m/s and 2.5 rad. A **0** in some scan cell (like 0/50) just means that picture's echoes failed round 2 or 3 — the earlier rounds still matched fine. The 0.50k-style totals (e.g. 0 matched of ~50k compared) are round-3 failures, not missing pictures.

So when FL shows scan numbers matching but detections at 0%: the photos were taken at the same moments, but the replay put **different cars at different distances** in them — while pointing the **same directions**.

4 · PhD-level layer analysis (FL)

Layer 1, scan alignment: DET scans IN/OUT = 395/378; vehicle/sim scans 395/378; common/input 372 / 395; processed/with-matches 372 / 371. Verdict: **healthy** (~95% overlap).

Layer 2, RDD prefilter (exact *rdd1_rindx, rdd1_dindx* equality, O(N) hash maps both sides): 373/374 common scans carry shared pairs; 62,250 pair comparisons entered layer 3 (0006; 46,764 for 0003). Verdict: **healthy** — the oft-suspected RDD stage is not the bottleneck.

Layer 3, threshold matrix (all four must pass):

signal	unit	|delta| p50	p90	p99	pass @2.5
ran	m	14.761	57.582	149.500	22.9%

signal	unit	|delta| p50	p90	p99	pass @2.5
vel	m/s	5.897	24.226	29.921	34.5%
theta	rad	0.457	1.770	2.301	99.9%
phi	rad	0.063	0.224	0.414	100.0%

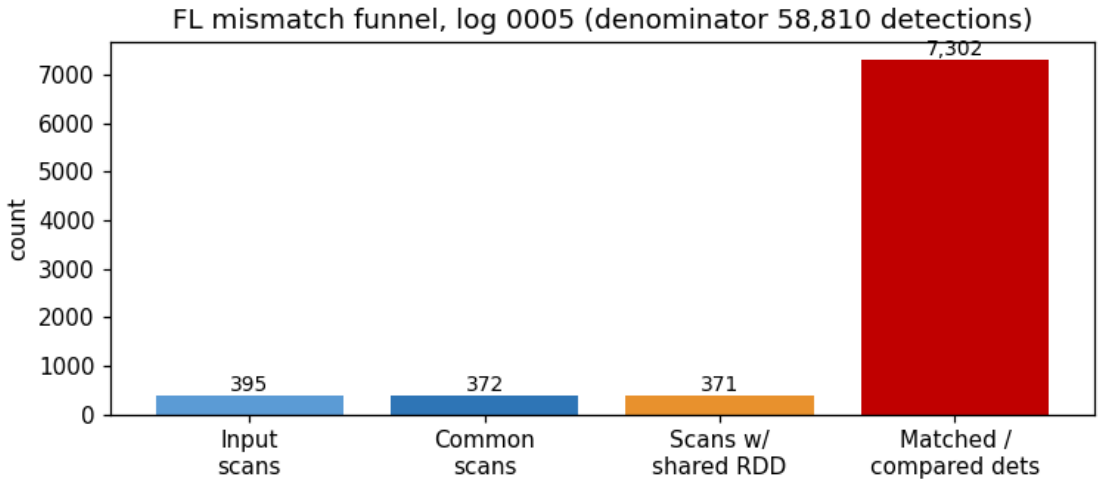
Reading: **theta 99.9% / phi 100%** — direction is preserved almost perfectly. **Range p50 ≈ 10.4 m** (only ~23–25% within 2.5 m), **velocity p50 ≈ 4.4–5.8 m/s** (~31–39% within 2.5 m/s). The joint pass rate is the product-like conjunction actually measured: **12.42% (7302/58810 matched)**. The delta spread (p50 10 m → p99 ~100 m) rules out a constant bias or a unit slip, both of which would give a tight shifted distribution; the content genuinely differs per echo while bearings agree. Interpretation: the resim replays similar scenes from similar angles with different traffic placement/speeds — direction-consistent, range/velocity-divergent.

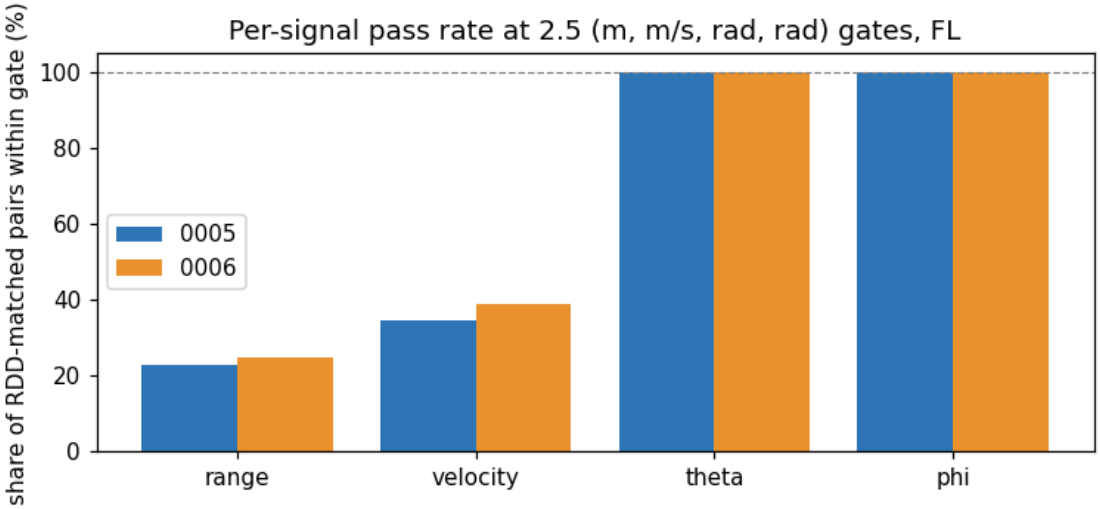
The four logs sit on one ladder (8.6% → 11.8% → 12.4% → 14.9% across 0003–0006) with the same fingerprint: healthy scans, healthy RDD overlap, loss concentrated in range, then velocity, angles near-perfect. 0004 shows the widest range spread (p50 17.4 m).

5 · Why alignment & tracker read as they do here

Alignment **failed** (not 0%): these AL logs contain no alignment stream at all — neither *DYNAMIC_ALIGNMENT_STREAM* nor *Dyn_Align_STREAM* — so there are no boresight signals to compare. Tracker **missing**: the *data/* AL logs carry *DETECTION_STREAM* instead of *OBJECT_LIST_STREAM* (only the separate *tracker/0003* log has object tracks), so there are no track IDs to index. Both are honest absences with stream inventories on the index page, not bugs.

6 · Figures





Per-signal pass rates at the 2.5 gates, this log (blue) vs 0006 (orange): angles ~100%, velocity ~31–39%, range ~22–25%. The headline is their conjunction.

7 · Recommendations

1) If the resim is *supposed* to reproduce traffic: investigate scenario seeding/traffic placement — range/velocity diverge while bearings hold. 2) If it is a *different* scenario by design: the 8–17% comparable-scan scores are expected; quote them with the coverage-inclusive overall beside them. 3) Gate changes will not fix this (p50 range ≈ 4× the gate); only scenario fidelity will. 4) For tracker-bearing AL logs use *tracker/0003*-style files with *OBJECT_LIST_STREAM*; for alignment, logs must include a Dyn_Align stream.